POWER BI PROJECT REPORT

Sales, Profit, Customer and Delivery Performance Analysis



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1. INTRODUCTION

1.1 Background

Shopora is an e-commerce business that sells products across different categories and serves customers in various regions. Understanding its sales performance, customer behavior, product profitability, and delivery operations is important for making effective business decisions.

This project uses data analysis and visualization to examine Shopora's business performance and identify areas of strength and opportunities for improvement.

1.2 Aim of the Project

The main aim of this project is to analyze Shopora's business data using Microsoft Power BI to generate meaningful insights and provide recommendations that can support better business decisions.

1.3 Project Objectives

The key objectives of this project are to:

- Analyze revenue and profit across product categories and subcategories.
- Identify monthly revenue and profit trends.
- Evaluate the effectiveness of discounting strategies.
- Examine product returns and their associated costs.

- Assess delivery performance and its potential relationship with customer behavior.
- Compare customer segments based on revenue and profitability.
- Identify profitable and loss-making product subcategories.
- Evaluate payment method and regional sales performance.
- Develop interactive dashboards and visualizations to communicate the findings.
- Provide practical recommendations to support Shopora's business growth and operational improvement.

1.4 Tools Used

Microsoft Power BI was used to prepare the data, create visualizations, analyze business performance, and develop interactive dashboards. Microsoft PowerPoint was used to present the key findings, insights, and recommendations.

2. DATA PREPARATION AND CLEANING

2.1 Dataset Overview

The original dataset consisted of four Excel tables containing order transactions, order items, customer information, and product details.

Table	Original records	Cleaned records
Orders	15,042	15,000
Order_Items	29,812	29,694
Customers	4,215	4,200
Products	314	309
Total	49,383	49,203

2.2 Data Cleaning Activities

The comparison identified the following changes:

- Duplicate removal: The cleaned files contain no fully duplicated rows. The original data had 42 duplicate rows in Orders, 118 in Order_Items, 14 in Customers, and 5 in Products. One additional customer record was also removed because it shared a Customer_ID with another record but had a different signup date.
- Missing values: The Orders table contains missing actual delivery dates, while the Order_Items table contains many missing return reasons. These missing

values remain in the cleaned dataset. They may be appropriate where an order has not been delivered or an item has not been returned.

- Data type correction: The Actual_Delivery_Date field in Orders and the Signup_Date field in Customers were converted to date formats. This supports accurate date-based analysis, such as monthly revenue trends and delivery performance.
- Data structure: The four tables and their column names were retained, allowing the dataset to be used for analysis across orders, products, and customers.

2.3 Importance of Data Cleaning

Data cleaning is an important stage of the analysis because duplicate records can inflate totals and lead to inaccurate business insights. Correct data types also help ensure that dates, numerical values, and other fields are analyzed appropriately.

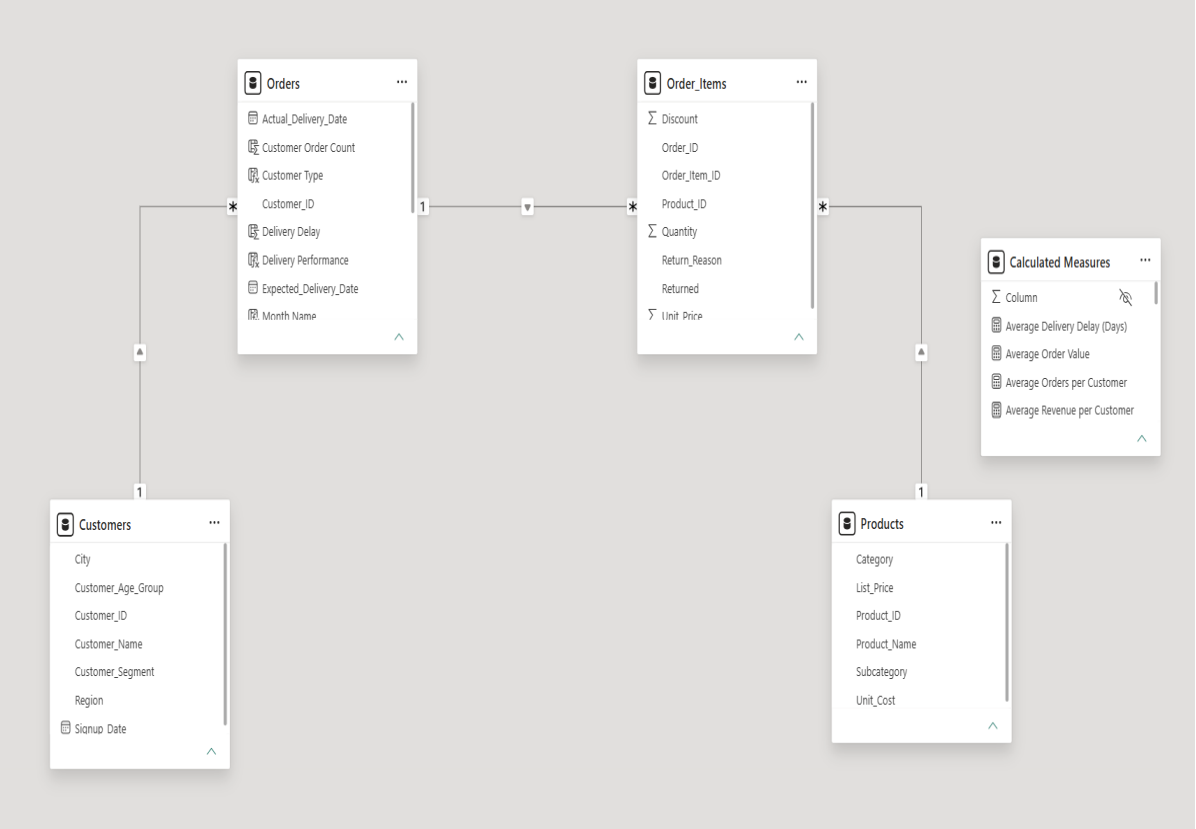
The cleaning process helped prepare Shopora's dataset for use in Power BI, where the data was used to develop visualizations, calculate performance measures, and answer the 10 business questions.

Note: This section is based on comparing the original and cleaned Excel files. The comparison shows the changes in the data, but it cannot confirm every individual Power Query action or the exact order in which you performed the cleaning steps.

2.2 Data Modelling

The data model was created in Power BI to connect the four main tables: Orders, Order_Items, Customers, and Products. Relationships were established using

common fields such as Customer_ID, Order_ID, and Product_ID. This structure allows data from different tables to work together for analysis and dashboard reporting. A separate Calculated Measures table was also created to organise key measures, including Average Delivery Delay, Average Order Value, Average Orders per Customer, and Average Revenue per Customer.



3. DASHBOARD DEVELOPMENT

3.1 Power BI Dashboard

Microsoft Power BI was used to transform the cleaned Shopora dataset into an interactive business intelligence dashboard. The dashboard was designed to present key business performance measures in a clear and accessible format and to allow users to explore the data through different business perspectives.

The dashboard combines key performance indicators (KPIs), charts, and filters to help users understand Shopora's revenue, profit, customers, products, delivery performance, and regional performance.

3.2 Dashboard Structure

The Power BI report was organized into several pages, with each page focusing on a particular area of the business.

- **Executive Dashboard**

  The Executive Dashboard provides a high-level overview of Shopora's performance. It presents key measures such as revenue, profit, customers, products, and other important business indicators.

  The page is designed to allow management to quickly identify the main strengths and issues within the business without needing to examine individual charts in detail.

- **Sales Performance**

  The Sales Performance page focuses on revenue and profit trends. It allows the performance of sales to be examined over time and supports comparisons between different periods.

- **Product Performance**

  The Product Performance page examines product categories and subcategories. It helps identify products that generate high revenue or profit as well as products that may be contributing to losses.

- **Customer Performance**

  The Customer Performance page compares different customer segments and customer types. It provides insight into customer value and purchasing behavior.

- **Delivery Performance**

  The Delivery Performance page examines whether Shopora is meeting its promised delivery dates. It includes measures such as late orders, late delivery rates, and delivery performance across customer groups.

- **Regional Performance**

  The Regional Performance page compares Shopora's sales performance across different regions and payment methods. This helps identify areas of stronger and weaker performance.

- **Detailed Analysis**

  A separate Detailed Analysis page was created to provide more detailed examination of the business questions. This page contains supporting visualizations that allow individual aspects of Shopora's performance to be explored in greater depth.

3.3 Interactivity

Interactive features were incorporated into the dashboard to allow users to explore the data dynamically. Filters and navigation elements help users move between different areas of the report and examine the information from different perspectives.

A page navigator was also used to provide consistent navigation between the main dashboard pages. The Detailed Analysis page was hidden from the main navigation while remaining available within the report when required.

3.4 Key Performance Measures

Several measures were created in Power BI to support the analysis. These included measures for revenue, profit, customers, products, subcategories, customer segments, and customer types.

For example, distinct counts were used to avoid counting the same customer or product multiple times. This helped ensure that KPIs represented the underlying business data accurately.

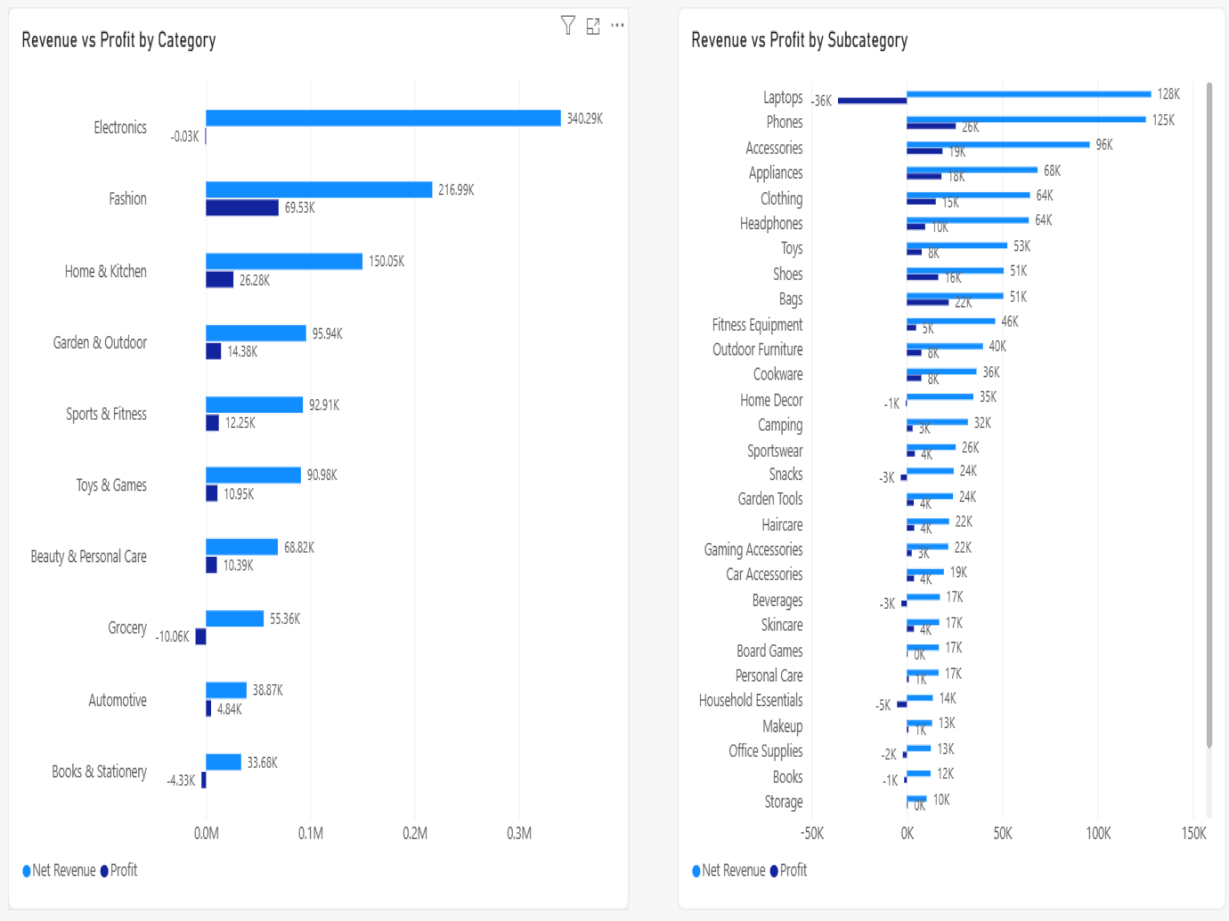
3.5 Purpose of the Dashboard

The overall purpose of the dashboard is to turn the cleaned Shopora dataset into useful business information. Rather than presenting the data as tables alone, the dashboard uses visualizations to highlight trends, differences, and potential areas for action.

The dashboard therefore acts as an interactive decision-support tool, while the accompanying PowerPoint presentation provides a focused explanation of the findings from the 10 business questions.

4. BUSINESS QUESTIONS AND FINDINGS

4.1 Question 1: Which Categories and Subcategories Generate the Most Revenue and the Most Profit?



Analysis

This question examines Shopora's revenue and profit across product categories and subcategories. The purpose is to identify which products generate the most sales income and which contribute the most to the business's profitability.

Key Findings

- Electronics generates the highest category revenue, at approximately 340.29K, but records a slightly negative profit of around -0.03K.
- Fashion generates approximately 216.99K in revenue and 69.53K in profit, making it the strongest category in terms of profit.
- Laptops generate the highest subcategory revenue, at approximately 128K, but record a loss of around 36K.
- Phones generate approximately 125K in revenue and 26K in profit.
- Accessories generate approximately 96K in revenue and 19K in profit.

Business Implications

The findings show that high revenue does not necessarily translate into high profit. Electronics generates the most revenue but contributes almost no profit, while Fashion generates less revenue but delivers considerably higher profit.

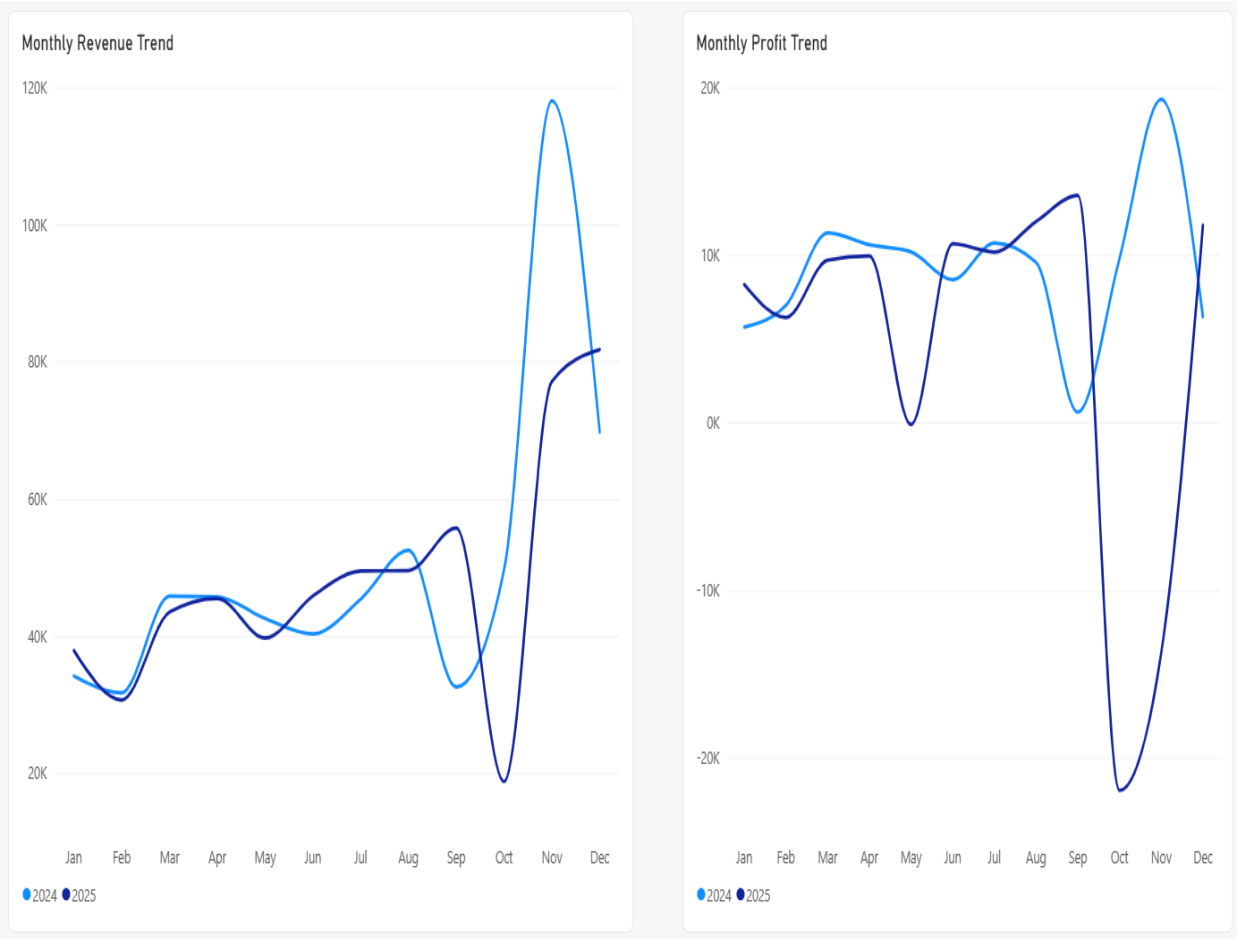
This difference highlights the importance of considering costs and profit margins when evaluating business performance.

Recommendations

- Investigate the costs, pricing, and discounting strategies associated with Electronics and Laptops.
- Review the successful profitability of Fashion to identify strategies that could be applied to other categories.
- Monitor the profit margins of high-revenue subcategories.

- Use both revenue and profit measures when making product and inventory decisions.

4.2 Question 2: Build a Monthly Revenue and Profit Trend Across Both Years. What Is Seasonal, and What Is Genuine Growth?



Analysis

This question examines Shopora's monthly revenue and profit trends across 2024 and 2025. The purpose is to identify periods of high and low performance, recognize possible seasonal patterns, and distinguish temporary sales increases from sustained business growth.

- Revenue remains relatively stable between approximately 30K and 55K during most months in both years.

- **November 2024** records a major revenue peak of approximately **118K**, followed by a decrease to around **70K in December**.
- In 2025, revenue drops to approximately **19K in October**, before recovering to around **77K in November** and **82K in December**.
- Profit also fluctuates throughout the two years, with a particularly noticeable decline in October 2025, when it falls to approximately **-22K**.
- The revenue increases toward the end of the year suggest possible seasonal patterns. However, the two-year comparison does not establish whether these patterns will recur.

Business Implications

The findings indicate that Shopora experiences significant monthly fluctuations rather than consistent growth throughout the year. The revenue peak in November 2024 may reflect seasonal demand, promotional activity, or other temporary factors.

The sharp revenue and profit decline in October 2025 is also important because it may indicate a period of weaker sales performance or increased costs. Comparing the same months across years is useful for understanding performance, but a longer historical record is needed to confirm genuine growth and recurring seasonality.

Recommendations

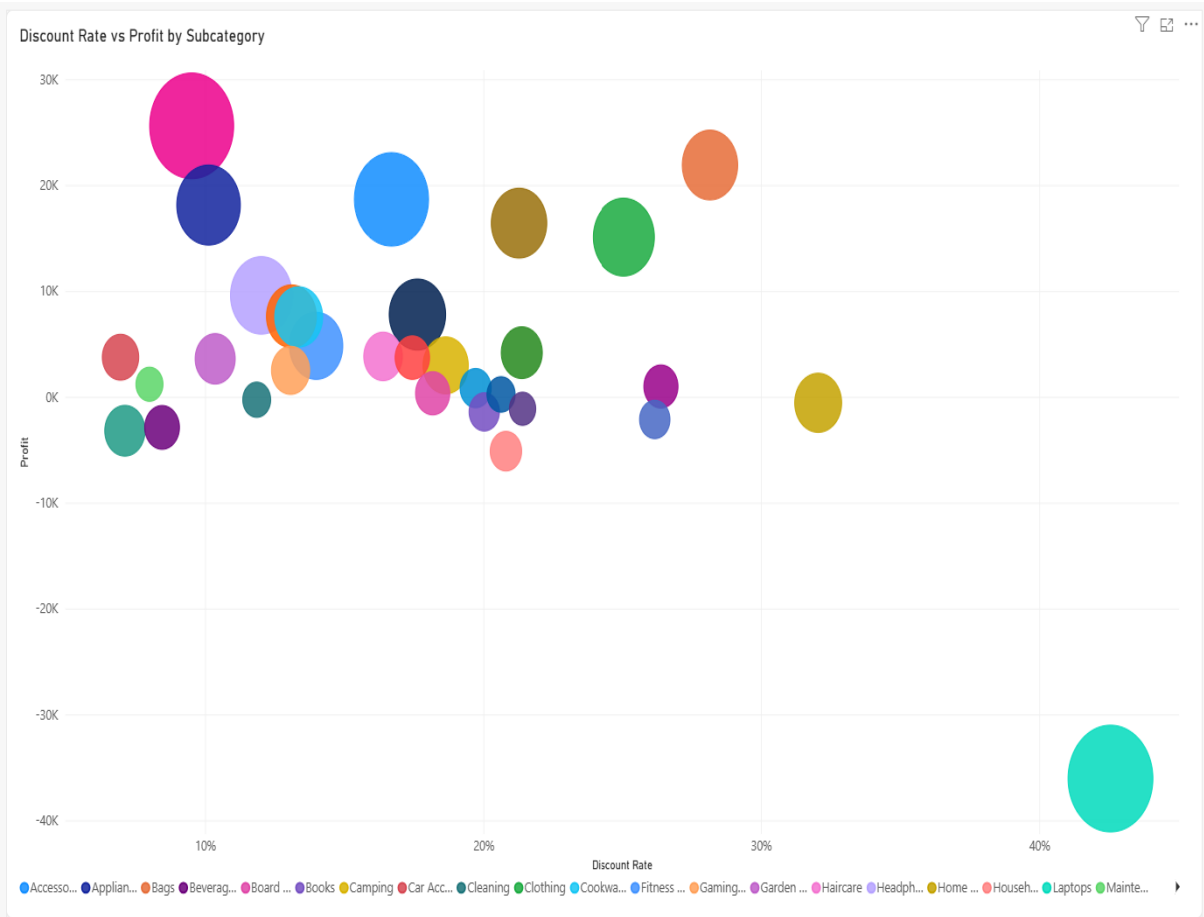
- Investigate the reasons behind the November 2024 revenue peak and determine whether it was associated with seasonal demand or promotional campaigns.

- Examine the October 2025 decline in revenue and profit to identify possible sales or operational issues.
- Compare monthly performance across additional years to establish reliable seasonal patterns.
- Plan inventory, staffing, and marketing activities around confirmed periods of high and low demand.
- Monitor both revenue and profit to ensure that sales increases translate into improved business performance.

4.3 Question 3: Analyse Shopora's Discounting. Is It Working?

Analysis

This question examines the relationship between discount rates and profit across Shopora's product subcategories. The objective is to understand whether discounting supports profitability or reduces the amount of profit the business earns.



Key Findings

- The chart shows considerable variation in profit across subcategories at different discount rates.

- Some subcategories achieve profits of approximately **26K** at discount rates around **10%–15%**.
- Other subcategories remain profitable at discount rates of approximately **20%–30%**, with one reaching around **22K** in profit.
- A subcategory with a discount rate of approximately **42%** records a loss of around **36K**.
- The results suggest that higher discount rates do not consistently lead to higher profits.

Business Implications

The findings indicate that discounting may be beneficial for certain product subcategories but may also reduce profitability in others. A high discount can help attract customers and encourage purchases, but it also reduces the revenue earned per unit sold.

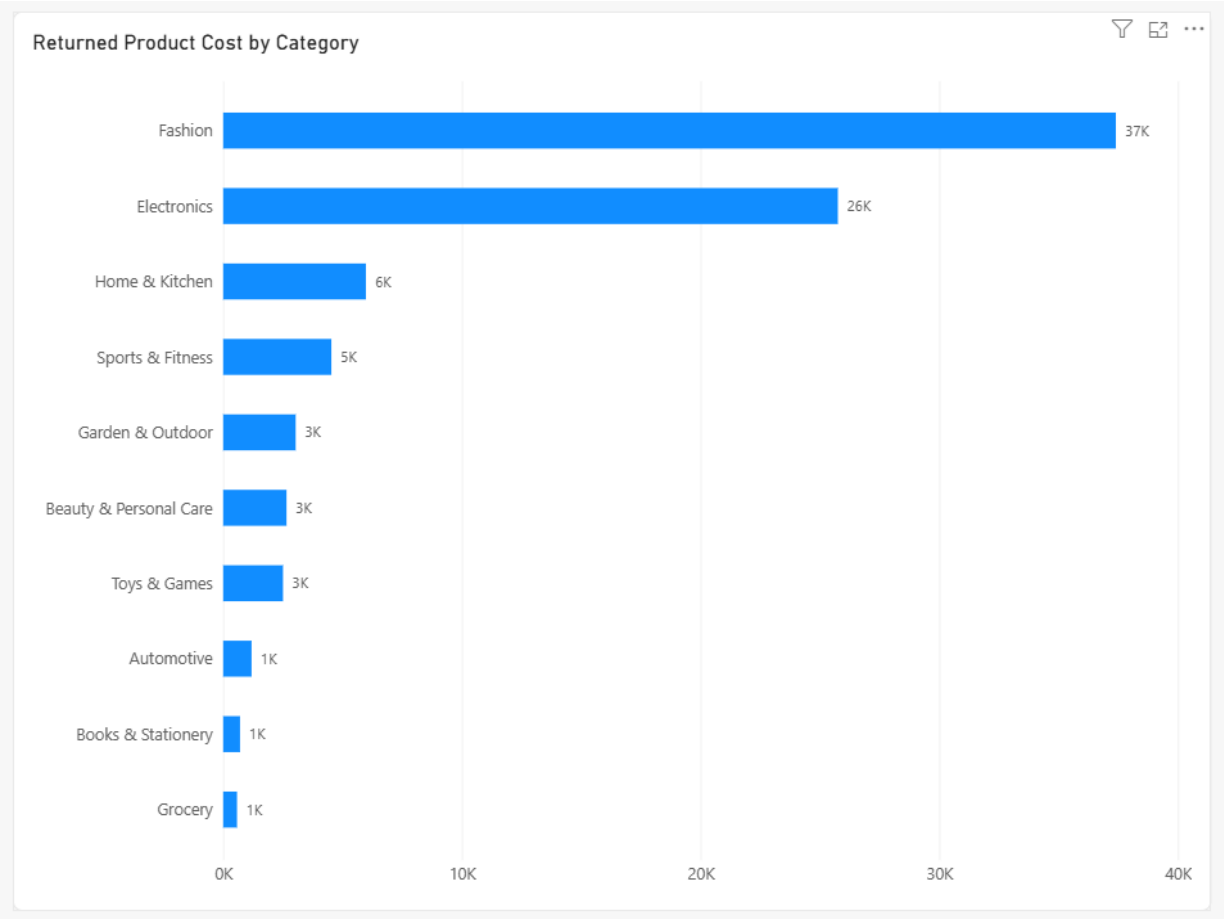
The chart alone cannot establish whether discounting is effective overall because it does not show what sales and profits would have been without discounts. Other factors, such as product costs and sales volume, may also influence profitability.

Recommendations

- Review subcategories with high discount rates and negative profits to identify potential pricing problems.

- Identify products that remain profitable at moderate discount rates and assess whether their strategies can be applied elsewhere.
- Establish discount limits based on product costs and desired profit margins.
- Compare sales volume and profit before, during, and after promotions to evaluate their effectiveness.
- Monitor discount performance by subcategory instead of applying a uniform discount strategy across all products.

4.4 Question 4: What Are Returns Costing the Business, and Where Are They Concentrated?



Analysis

This question examines the cost of returned products across Shopora's product categories. The objective is to identify which categories contribute most to return-related costs and where the business should focus its efforts to reduce losses.

- **Fashion** has the highest returned product cost, at approximately **37K**, making it the largest contributor among the categories shown.
- **Electronics** is the second-highest category, with returned product costs of approximately **26K**.

- **Home & Kitchen** accounts for approximately **6K**, while **Sports & Fitness** contributes around **5K**.
- **Garden & Outdoor, Beauty & Personal Care, and Toys & Games** each have returned product costs of approximately **3K**.
- **Automotive, Books & Stationery, and Grocery** have the lowest returned product costs, at around **1K** each.
- Returns are heavily concentrated in Fashion and Electronics compared with the other categories.

Business Implications

The findings suggest that Fashion and Electronics are the main areas contributing to Shopora's returned product costs. High return costs can reduce profitability and may indicate issues with product quality, customer expectations, product descriptions, or sizing.

However, returned product cost alone does not show which categories have the highest return rates. Categories with greater sales volumes may naturally have higher return costs.

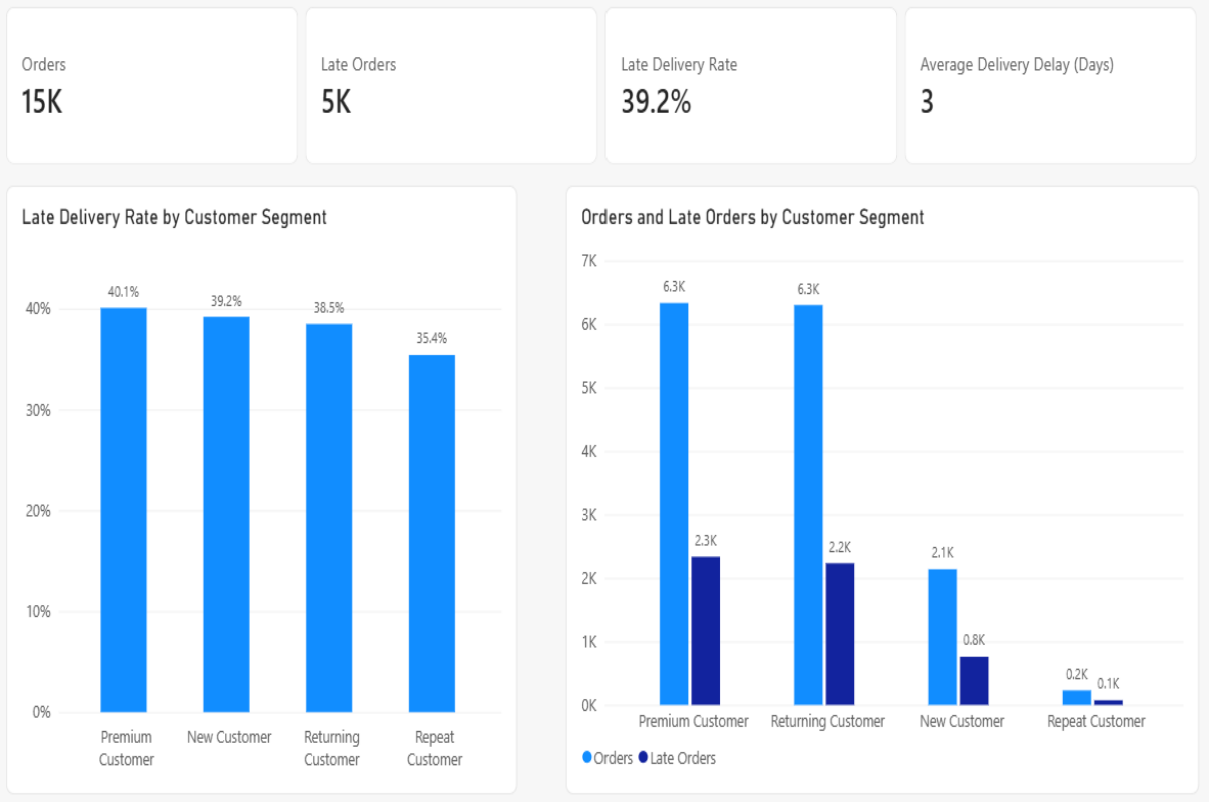
Recommendations

- Investigate the reasons for returns in Fashion, particularly sizing issues, product descriptions, and differences between product images and actual products.

- Review product quality, specifications, and customer expectations in Electronics.
- Analyze return reasons and rates within each category to identify specific products contributing most to the costs.
- Improve product descriptions, sizing guides, and quality checks where recurring problems are identified.
- Monitor return costs and return rates regularly to evaluate the effectiveness of improvements.

Note: The chart measures returned product cost, not the number of returned items or the return rate. Additional sales and returns data would be needed to make a fair comparison of return frequency across categories.

4.5 Question 5: How Well Does Shopora Keep Its Delivery Promise, and Does It Matter?



Analysis

This question evaluates Shopora's delivery performance by examining late delivery rates, average delivery delays, and differences in delivery performance across customer segments. The objective is to determine how reliably Shopora meets its promised delivery dates and identify areas that require operational improvement.

Key Findings

- Shopora has approximately **15,000 orders**, of which around **5,000 are late**.

- The overall late delivery rate is approximately **39.2%**, meaning that nearly four out of every ten orders arrive late.
- The average delivery delay is approximately **3 days**.
- **Premium Customers** have the highest late delivery rate at **40.1%**, followed by New Customers at **39.2%**.
- **Returning Customers** have a late delivery rate of **38.5%**, while Repeat Customers have the lowest rate at **35.4%**.
- Premium and Returning Customers each account for approximately **6,300 orders**, with around **2,300 and 2,200 late orders**, respectively.

Business Implications

The high late delivery rate suggests that Shopora faces challenges in meeting its promised delivery dates. Delays can affect customer satisfaction, trust, and the overall shopping experience.

The differences in late delivery rates across customer segments also highlight the importance of monitoring delivery performance for different groups. Premium Customers have the highest late delivery rate, which is a concern because they contribute significantly to the company's revenue and profit.

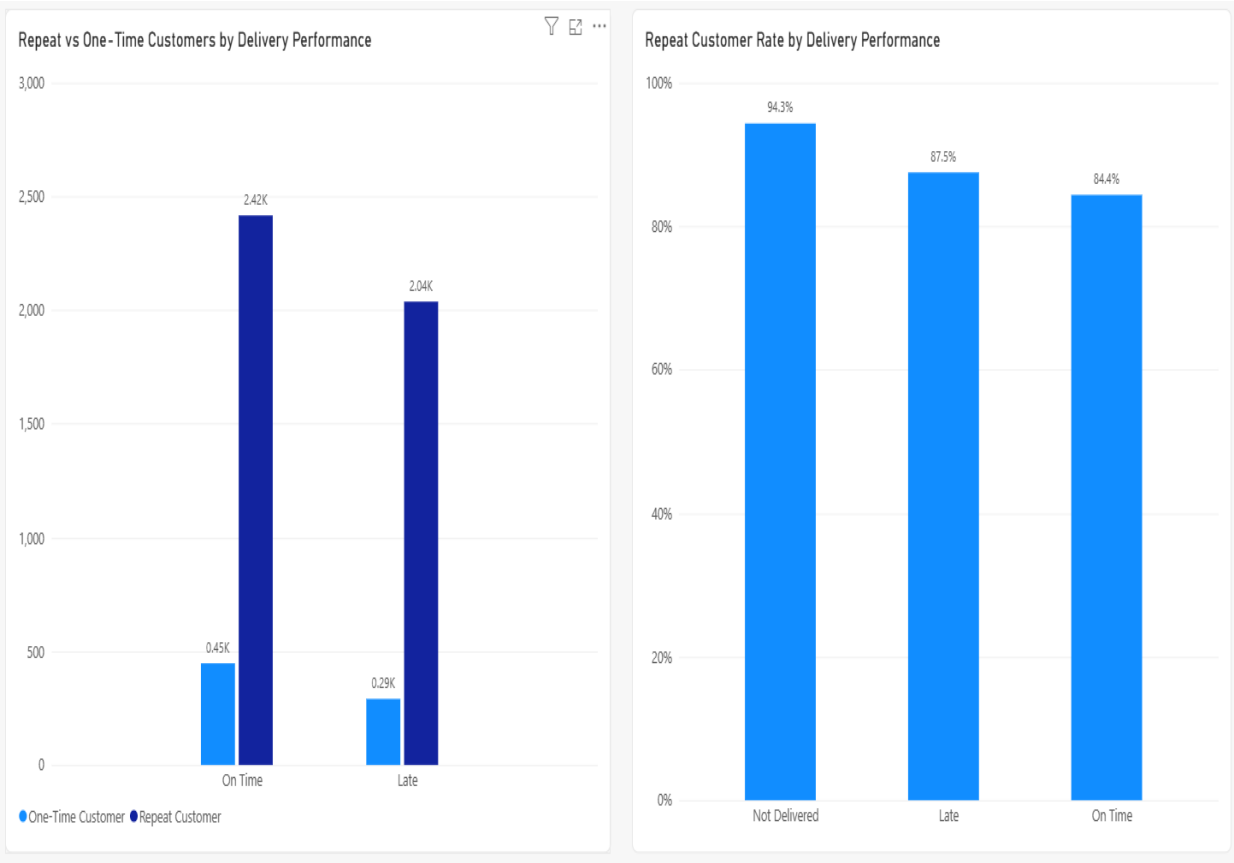
However, the charts do not directly establish how late deliveries affect customer satisfaction or future purchasing decisions.

Recommendations

- Investigate the causes of late deliveries, particularly among Premium Customers.
- Review shipping processes and delivery schedules to identify bottlenecks and reduce delays.
- Monitor late delivery rates and average delivery delays regularly across customer segments.
- Consider improving delivery reliability for high-value customers while maintaining consistent service for all customers.
- Compare customer satisfaction and repeat purchase behavior with delivery performance to understand the broader business impact.

Key takeaway: Shopora's late delivery rate of approximately 39.2% indicates a significant operational challenge. Improving delivery reliability could strengthen the customer experience, but further analysis is needed to measure its effect on customer loyalty.

4.6 Question 6: Does Anything in the Data Suggest Late Delivery Affects Customer Behaviour?



Analysis

The dashboard compares repeat and one-time customers across delivery performance categories: on-time, late, and not delivered. It also shows the repeat customer rate for each delivery category.

The charts indicate that:

- For on-time deliveries, there were approximately 2,420 repeat customers and 450 one-time customers.

- For late deliveries, there were approximately 2,040 repeat customers and 290 one-time customers.
- The repeat customer rate was 84.4% for on-time deliveries and 87.5% for late deliveries.
- The not-delivered category showed a repeat customer rate of 94.3%, which is notably higher than the other two categories.

Key Findings

The dashboard does not show a lower repeat customer rate for late deliveries. In fact, the displayed rate is slightly higher for late deliveries than for on-time deliveries. This suggests that late delivery is not clearly associated with reduced repeat purchasing in the results shown.

However, the figures do not prove that late delivery improves customer loyalty or has no negative effect. The dashboard shows an association, not a causal relationship. The particularly high rate for the not-delivered category also deserves further investigation, including checking how that category is defined and which orders or customers are included.

Recommendations

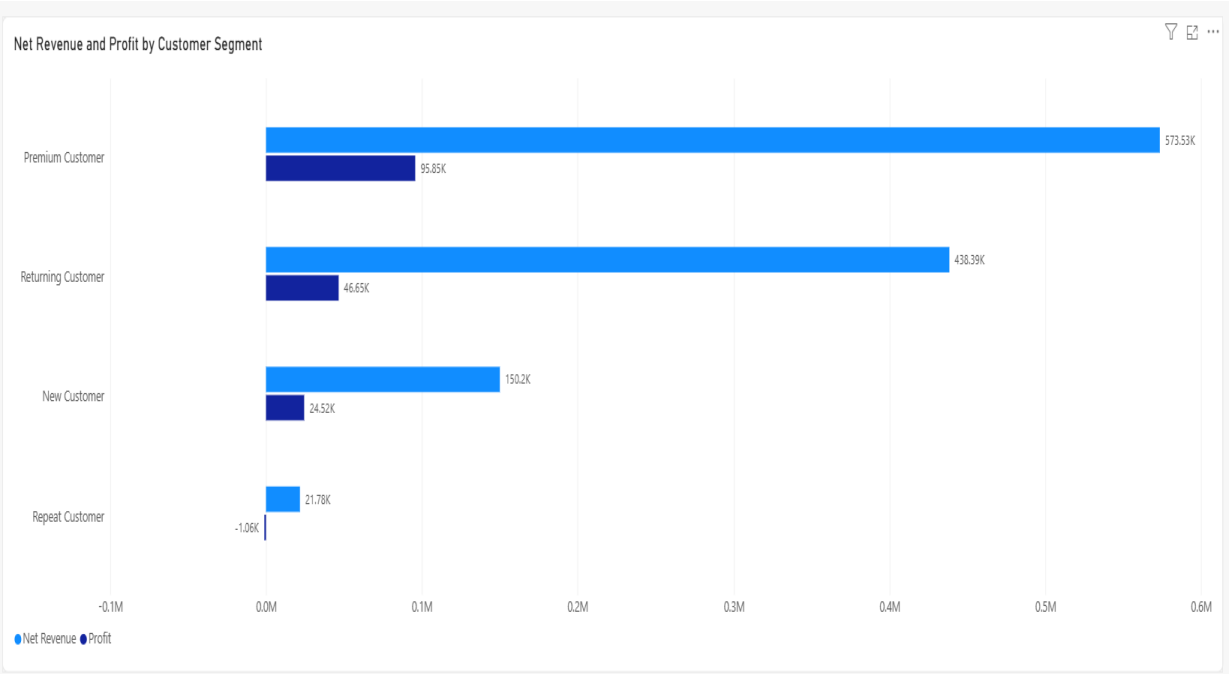
- Compare customer purchasing behaviour over time to see whether customers who experience late deliveries place fewer subsequent orders.
- Investigate the not-delivered category to confirm that its records and repeat-customer calculation are being interpreted correctly.

- Where possible, compare customer satisfaction, complaints, and repeat purchases alongside delivery performance.
- Avoid concluding that late deliveries have no impact based on these charts alone.

Conclusion

The available dashboard results do not suggest that late deliveries are associated with a lower repeat customer rate. The late-delivery group has a slightly higher displayed rate than the on-time group. Further customer-level and time-based analysis is needed to determine whether delivery delays affect future purchasing behaviour.

4.7 Question 7: Which Customer Segments Contribute the Most Revenue and Profit?



Customer Segment Performance and Late Delivery Rate

Customer_Segment	Orders	Late Delivery Rate
New Customer	2138	39.2%
Premium Customer	6333	40.1%
Repeat Customer	229	35.4%
Returning Customer	6300	38.5%
Total	**15000**	**39.2%**

Analysis

The dashboard compares Shopora's customer segments by revenue, profit, and order volume. The results show differences in the financial contribution of Premium, Returning, New, and Repeat customers.

The charts indicate that:

- Premium customers generated the highest revenue, approximately 573.53K, and the highest profit, approximately 95.85K.

- Returning customers generated approximately 438.39K in revenue and 46.65K in profit.
- New customers contributed approximately 150.20K in revenue and 24.52K in profit.
- The dashboard shows approximately 6,333 orders from Premium customers and 6,300 from Returning customers.
- The Repeat segment displayed a small negative profit of approximately -1.06K.

Key Findings

Premium customers are the strongest contributors to both revenue and profit among the segments shown. Returning customers also make a substantial contribution, with revenue and profit below the Premium segment.

New customers contribute less overall revenue and profit, which may reflect their smaller contribution to order volume. The negative profit shown for the Repeat segment is worth investigating, as it may indicate that this group generates sales without equivalent profitability.

The dashboard uses both “Returning” and “Repeat” as segment labels. Since their precise definitions are not established by the displayed results, they should not automatically be treated as interchangeable.

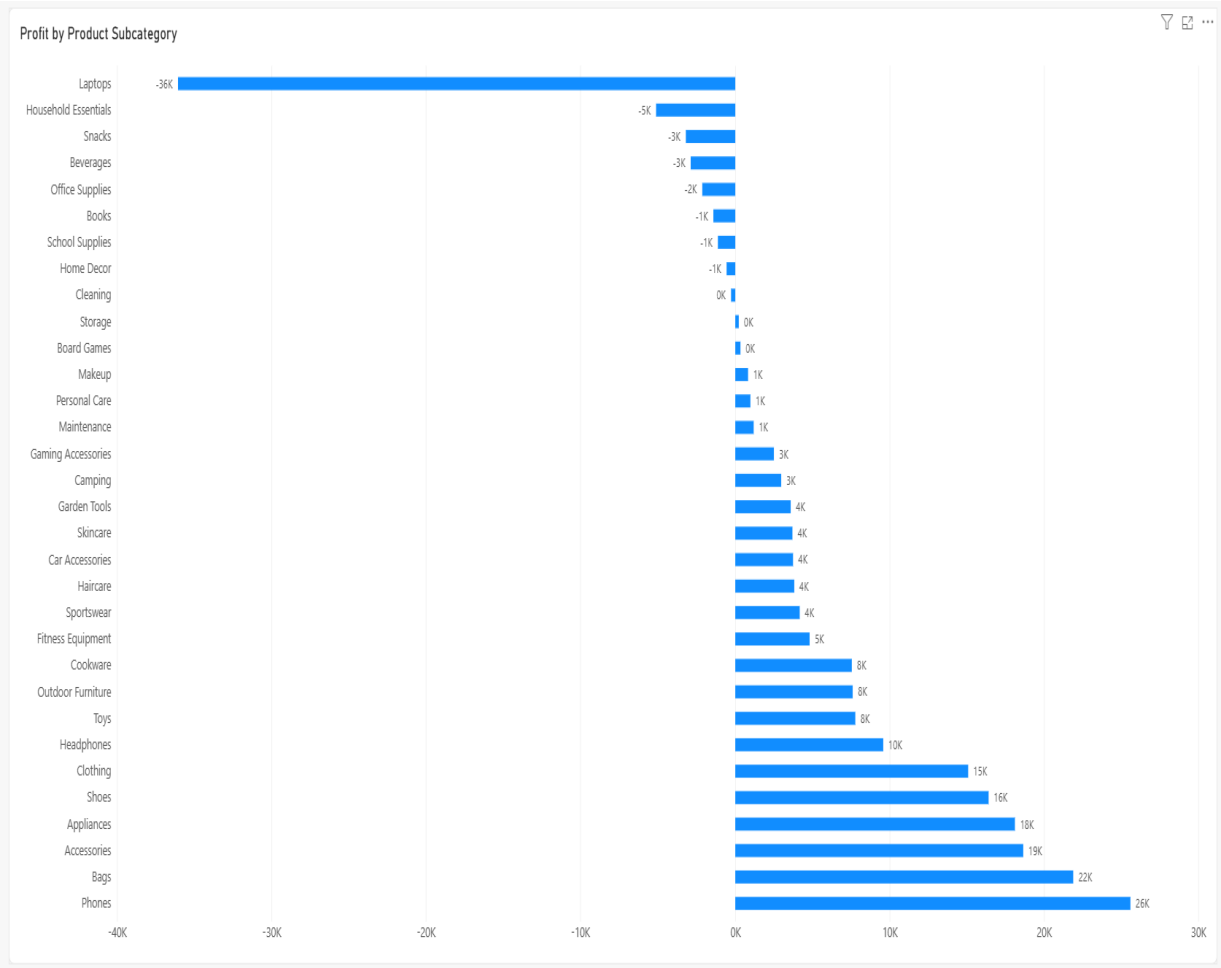
Recommendations

- Prioritise customer retention strategies for Premium customers, given their strong contribution to revenue and profit.
- Encourage New customers to make additional purchases through relevant follow-up offers and product recommendations.
- Investigate the negative profit shown for the Repeat segment, including its discounts, product mix, and order costs.
- Review the definitions of the Returning and Repeat segments to ensure that customer groups are clearly distinguished in future analysis.

Conclusion

Premium customers contribute the most revenue and profit in the dashboard results, while Returning customers are another important source of business performance. The negative profit for the Repeat segment highlights an area for further investigation. Clear segment definitions and additional analysis could help Shopora make more targeted and profitable customer decisions.

4.8 Question 8: Are There Products or Subcategories the Business Would Be Better Off Without?



Analysis

The dashboard uses a bar chart titled Profit by Product Subcategory to compare the profitability of Shopora's product subcategories. The chart highlights which subcategories generate positive profit and which are operating at a loss.

The results show that:

- Laptops generate the largest loss, at approximately -36K.
- Household Essentials show a loss of approximately -5K.

- Snacks and Beverages each show losses of around -3K.
- Office Supplies show a loss of approximately -2K.
- Several other subcategories, including Books, School Supplies, and Home Decor, have smaller losses.
- Phones generate the highest profit, at approximately 26K.
- Bags contribute around 22K, while Accessories generate approximately 19K.
- Appliances and Shoes are also strong contributors, with profits of approximately 18K and 16K, respectively.

Key Findings

Laptops stand out as the most unprofitable subcategory in the chart, with a substantial loss compared with all other categories. Household Essentials, Snacks, and Beverages also show negative profit, although their losses are considerably smaller.

In contrast, Phones, Bags, Accessories, and Appliances are among the strongest profit contributors. This indicates that Shopora's profitability varies significantly across its product range.

However, a negative profit figure alone does not prove that the business would be better off discontinuing a subcategory. The chart does not show sales volume, customer demand, strategic importance, or whether the losses could be reduced through pricing or cost changes.

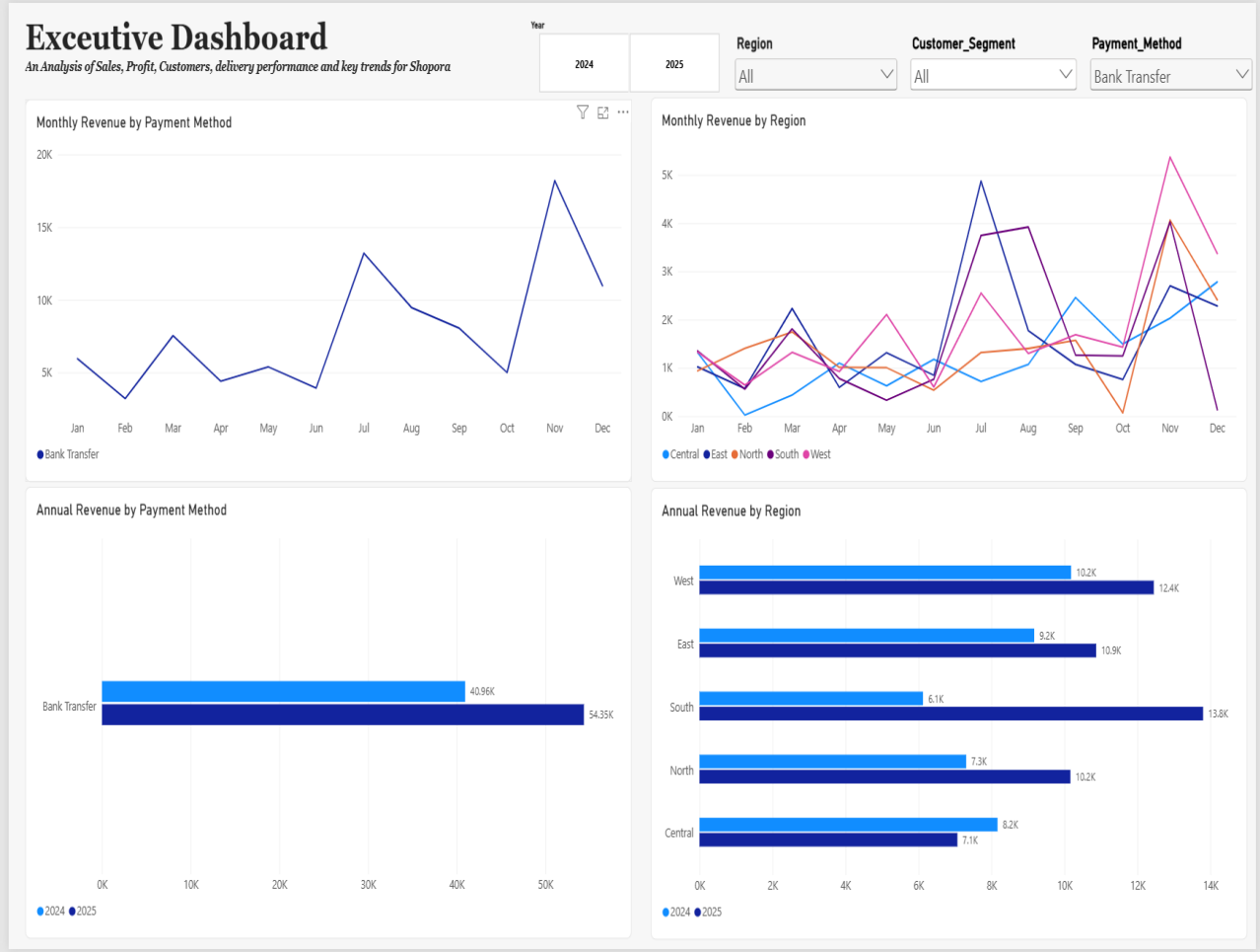
Recommendations

- Investigate laptops first, reviewing their pricing, supplier costs, discounts, and return-related expenses to identify the cause of the large loss.
- Review the smaller loss-making subcategories to determine whether their profitability can be improved.
- Consider pricing adjustments, supplier negotiations, or more selective promotions before deciding to discontinue products.
- Continue supporting strong profit contributors such as Phones, Bags, Accessories, and Appliances.
- Assess customer demand and the wider product range before removing any subcategory, as some products may support sales of other items.

Conclusion

The profit analysis identifies Laptops as the clearest area of concern, with a loss of approximately 36K. Household Essentials, Snacks, and Beverages also show negative profit. Shopora should investigate these subcategories and explore ways to improve their performance before considering discontinuation. The chart provides a useful starting point for product decisions, but further analysis is needed to determine whether removing any subcategory would benefit the business.

4.9 Question 9: Which Payment Methods and Regions Are Growing, and Which Are Not?



Analysis

The executive dashboard compares Shopora's monthly and annual revenue by payment method and region for 2024 and 2025. These charts help identify changes in revenue and highlight which regions and payment methods are performing better.

The dashboard shows the following results:

- Bank Transfer: Annual revenue increased from approximately 40.96K in 2024 to 54.35K in 2025, indicating growth of around 13.39K.
- West: Revenue increased from approximately 10.2K in 2024 to 12.4K in 2025.
- East: Revenue increased from approximately 9.2K to 10.9K.

- South: Revenue increased substantially, from approximately 6.1K to 13.8K, making it the strongest-growing region shown.
- North: Revenue increased from approximately 7.3K to 10.2K.
- Central: Revenue decreased from approximately 8.2K to 7.1K, making it the only region shown with a decline.

Key Findings

Bank Transfer revenue grew between 2024 and 2025, suggesting that this payment method contributed more revenue in the later year. However, the dashboard only displays Bank Transfer in the payment-method charts, so it does not support a comparison with other payment methods.

The regional results show growth in four out of five regions. The South experienced the largest increase, while the West remained the highest-revenue region in 2025. Central was the only region with lower annual revenue in 2025 than in 2024.

The monthly charts also show fluctuations throughout the year. For example, Bank Transfer revenue peaks in November, while regional revenue peaks vary across months. These patterns suggest that revenue performance changes over time rather than increasing steadily every month.

Recommendations

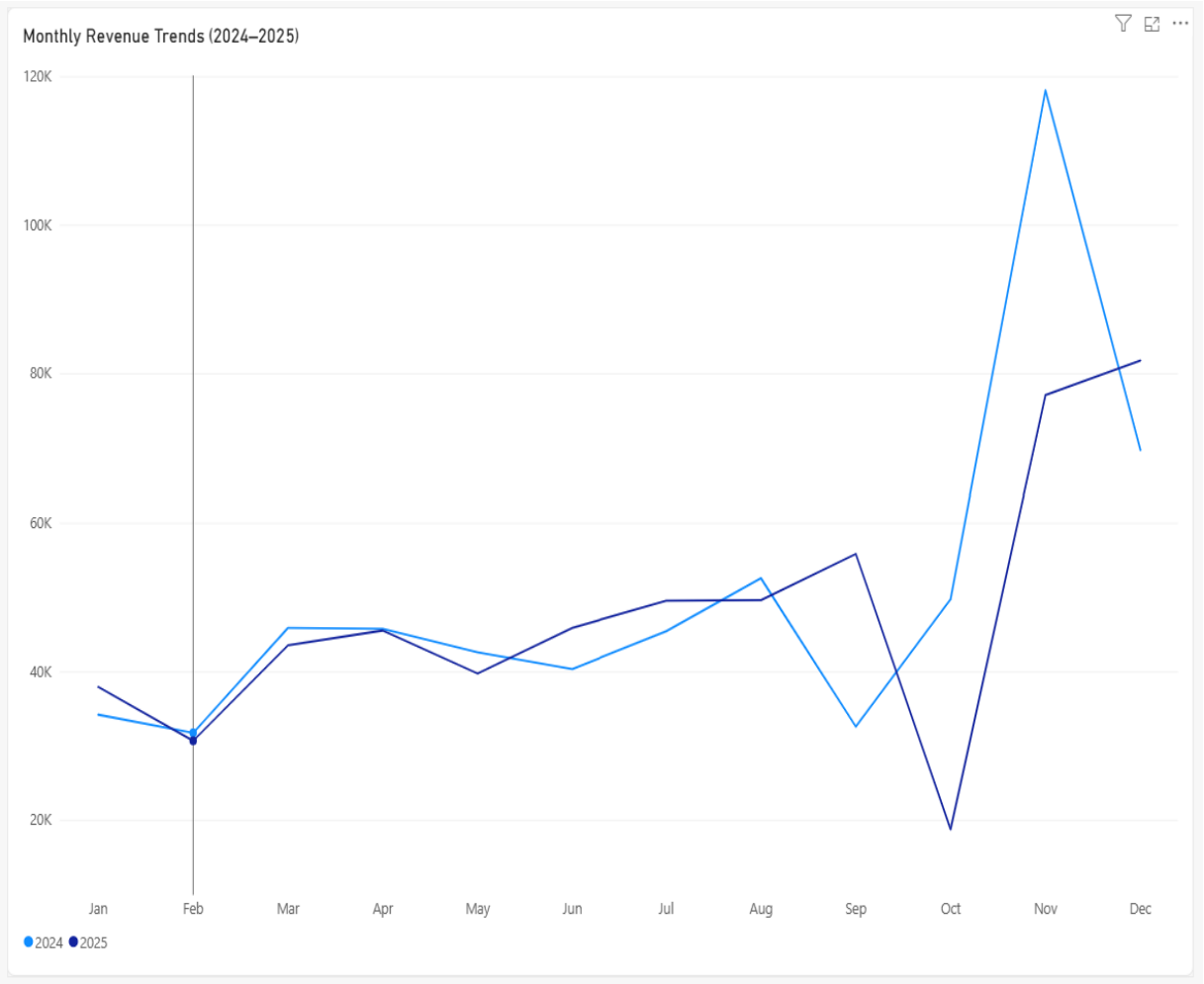
- Build on growth in the South by examining which products, customer segments, or sales activities contributed to its strong increase.

- Maintain the West's strong performance while investigating opportunities to sustain its revenue growth.
- Review Central's decline to identify possible causes, such as changes in demand, product mix, or customer activity.
- Continue supporting Bank Transfer while comparing it with other payment methods in a more complete payment-method analysis.
- Monitor monthly revenue patterns to help plan promotions, inventory, and regional sales activities around stronger and weaker periods.

Conclusion

The dashboard indicates that Bank Transfer revenue increased from 2024 to 2025. Regionally, the South showed the strongest growth, followed by the North, West, and East. Central was the only region to experience a revenue decline. These findings can help Shopora focus on growing markets and investigate weaker performance. However, the payment-method comparison is limited because only Bank Transfer is displayed, and the annual figures alone do not explain the reasons behind the changes.

4.10 Question 10: Are There Particular Months When Shopora Generates the Most Revenue, and Why Does This Matter?



Analysis

The dashboard presents monthly revenue trends for 2024 and 2025. Comparing the two years helps identify months with higher revenue, periods of weaker performance, and possible seasonal patterns.

The chart shows the following trends:

- November 2024 was the highest-revenue month shown, with approximately 118K.

- December 2025 generated approximately 82K, the highest month in 2025.
- November 2025 also performed strongly, with approximately 77K in revenue.
- Revenue in most months from January to August remained within a range of approximately 30K–53K across the two years.
- October 2025 was the lowest month shown, with revenue of approximately 19K.
- Revenue increased sharply from October to November in both years, although the size of the increase differed considerably.

Key Findings

The chart suggests that November and December are important revenue months for Shopora. November 2024 showed a particularly large peak, while November and December 2025 also recorded strong revenue.

The sharp increases toward the end of the year may indicate seasonal demand or the influence of promotional activity. However, the chart alone cannot confirm why revenue increased. Further analysis would be needed to determine whether the peaks were driven by customer demand, discounts, marketing campaigns, or other factors.

The differences between 2024 and 2025 also show that monthly revenue is not consistent from one year to the next. October 2025, in particular, had much lower revenue than October 2024.

Recommendations

- Prepare for the November and December peak period by reviewing inventory, staffing, and delivery capacity in advance.
- Examine the sales and marketing activities that took place during high-revenue months to identify practices worth repeating.
- Investigate the sharp decline in October 2025 to understand what may have contributed to the weaker performance.
- Use monthly revenue trends to support sales forecasting and budget planning.
- Compare revenue with profit during peak months to ensure that higher sales also translate into stronger profitability.

Conclusion

The dashboard identifies November and December as important months for Shopora's revenue, with November 2024 showing the largest peak at approximately 118K. These patterns matter because understanding when revenue is strongest can help the business plan inventory, marketing, staffing, and delivery operations. However, further analysis is required to confirm the causes of these peaks and determine whether the patterns represent consistent seasonal behaviour.